מטלה 1 Project Initiation Document

בחרנו את הפרויקט שלנו מכיוון שהמון צעירים עוברים לעיתים תכופות יחסית לדירות, מעבר דירה ידוע כפעולה לא קלה הדורשת המון מאמץ והמון מוצאים את עצמם מסתבכים בתהליך מעבר לצפוי. מעבר דירה הינו תהליך הדורש סדר, ארגון ותכנון על מנת שיבוצע באופן חלק. לכן, חשבנו על אפליקציה שתקל על התהליך ותוכל להוות מעטפת להליך מעבר הדירה. באופן אישי, אחד מחברי הקבוצה עובר דירה בקרוב ומטרת האפליקציה למנוע קשיים ותקלות במהלך מעברו.

1. Business Case & Project Charter

- **Project Name**: EasyMove

- **Problem Statement**: The application aims to address the difficulties and challenges encountered during moving apartments.

- **Market Research**: Currently, there already exists an application named “הובלות בקוויק”, however, it does not offer much practical functionality apart from providing basic contact details of transport drivers, as of today (10th of November), the application contains the contact details of only 60 drivers, meaning, the app is does not try to innovate since there is virtually no competition in the market, so the developers seemingly don’t feel the need to put too much effort. We aim to introduce a bit more competition to the market, and to provide the first real apartment moving app.

- **Purpose and Objectives:** The project is being undertaken by us, due to the following reasons:

1. Many people struggle when moving to a new apartment, and, the application will provide users with a concrete and direct solution/framework for facilitating the process.
2. There is no real market, and as a consequence, no competition that really attempts to provide an intuitive and elegant software solution to this problem. Our goal is to provide such a solution.

- **Benefits and Impact:** The expected outcomes for this project are as follows:

1. Users of our application will have a better and smoother apartment moving experience.
2. Introduce more competition and interest in this particular industry.
3. Enhanced security; phone numbers will not be exposed during the moving process, users and transporters will communicate via secure means.

- **Scope:** The included features in our scope are as follows:

1. The ability to create records/lists of items that the user is interested in moving.
2. The ability to search for available transporters with high ratings in ones' area.
3. Secure chat functionality between the user and the transporter for easy communication during transports.
4. The ability for two people (mainly couples), to share the same transport process (same driver, same transportation company), in order to get a discount and save expenses.
5. Straightforward, intuitive and smooth user interface with an always present "burger" menu.
6. A transporter registered in the system must fill out an "about me" section.
7. Users can provide feedback/commentary about their experience with a specific transporter in order to better notify future clients about said transporter.

Possible boundaries/limitations we might face:

1. Real-time tracking of the transporter's location will not be available.
2. The estimated required size of the truck used for transportation can not be determined beforehand, which is why the transporter must communicate with the user to get this information. The same goes for the price of the transportation.
3. Payments will not be handled directly within the app.

- **High-Level Risks:** A few possible key risks:

1. The application is highly dependent on the transporters that register in the system in order to provide variety (truck size, etc.) for users with different needs. As a result of possible low variety of transporters, this might cause users to fallback to traditional moving methods.
2. The application is also highly dependent on the number of users registered, as a low amount of users might take away potential interest from interested transporters, and many transporters might have this profession as their main income, and it would affect them. Because of the nature of this industry's market, most users might download the application for one-time usage and forget about it once they have finished moving to a new apartment.

- **Alternative Solutions:** An alternative way is the traditional way that most people are familiar with, which involves mainly searching online for phone numbers, making their own lists, going through multiple different applications, hoops and hurdles to reach the same result less efficiently. Our application reduces the amount of steps that the user needs to do, and provides everything in one place.

- **Key Stakeholders:**

- Private end-users that want to move to a new apartment.
- Companies that want to move their resources.
- Movers that want to publish and operate their business through the app.
- Development team that is responsible for designing & developing the app.
- Product managers and marketing team for the app distribution.
- Legal advisors to protect the company.

- Resources: Provided resources:

- Android Studio (Java).
- Firebase (Google).
- Algorithmic & Programming Knowledge.
- Ariel University's Facilities, guidance from the lecturer and TAs.
- Trello (Task Management).
- GitHub (version control, project architecture & organization)

2. Statement of Work (SOW)

- Vision Statement: Our vision statement is to create a highly functional tool & framework for facilitating the moving process for users.

Project objectives to achieve:

1. Clean & friendly user interface, pleasant experience using the app.
2. To successfully realize the previously discussed features to be implemented.

- Scope of Project:

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1. Clean & friendly user interface, pleasant experience using the app.
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We state the boundaries of the project:

1. The application will be limited to the region of Israel, and will not have scalability for other countries/regions, due to different market natures, competitiveness in different regions.
2. We would say that one significant boundary is that we can not really audit the quality of the service & reliability of transporters.
3. Our application only provides a user-friendly platform, and is not able to supervise the methods of conduct for transporters. We cannot take accountability for any shortcomings or the reliability of transporters.
4. There is no in-app payment services. The transporter and users must agree on a way to perform transactions.

- Scope of Work: The included features in our scope are as follows:

1. The ability to create records/lists of items that the user is interested in moving.
2. The ability to search for available transporters with high ratings in ones' area.
3. Secure chat functionality between the user and the transporter for easy communication during transports.
4. The ability for two people (mainly couples), to share the same transport process (same driver, same transportation company), in order to get a discount and save expenses.

We would like to design a unique "vibe" or "feel" for the app to make it memorable. The development/testing/implementation phases would be done by the team members exclusively.

- Key Features: The included features in our scope are as follows:

1. The ability to create records/lists of items that the user is interested in moving.
2. The ability to search for available transporters with high ratings in ones' area.
3. Secure chat functionality between the user and the transporter for easy communication during transports.
4. The ability for two people (mainly couples), to share the same transport process (same driver, same transportation company), in order to get a discount and save expenses.

- Constraints: Possible constraints/limitations we might face:

1. Real-time tracking of the transporter's location will not be available.
2. The estimated required size of the truck used for transportation can not be determined beforehand, which is why the transporter must communicate with the user to get this information. The same goes for the price of the transportation.

- Dependencies: The project's dependencies are as followed:

- Android Studio (Java).
- Firebase (Google).
- GitHub (version control, project architecture & organization)
- Developers team (Backend, Full stack, Frontend)
- UI/UX designer
- Access to GPS and camera
- Server to store chats and data

- Deliverables: The project will produce documentation and the application itself (with all of it's already well known features).

- Timeline: End of the semester: 16/01/26

- Performance Criteria: We will preform extensive testing and also user simulations to get our feedback on the smoothness and responsiveness of the application and also make sure it operates as intended.

- Risk Management:

- Risk Identification: Currently our application is depended on FireBase service, which is known to have a free version to basic usage, but we are facing a risk that if our application is using the paid version that we would need to implement ad based revenue system and by that we are risking losing profits and users.

- Mitigation Strategies: We would try to avoid paid services as much as we can and if we must implement advertisements, we will try to keep the user experience as clean as possible.

- Contingency Plans: We would implement ad based revenue system.

3. Feasibility Study Report

- Technical Feasibility: The project is considered feasible for the following reasons:

- Android Studio is already a very comprehensive and full-featured development suite, it is already mature enough for providing a friendly development environment.
- All of the app's features were implemented multiple times in other different apps, in different circumstances.
- Many different applications rely on the same database management system (i.e. Firebase).

- Operational Feasibility: Companies and organizations often need an easy and cost effective logistic solution to moving products and resources (mainly to warehouses). Our application may provide a good framework to integrate into the company's operation methods.

- Financial Feasibility:

Cost	Estimated hours	Development phase
5000$	50	Characterization and definition of requirements
3000$	30	UX/UI design
15000$	150	Frontend development
20000$	200	Backend development
6000$	60	Integrations
7000$	70	QA Tests
5000$	50	Debugging
61000$	610	Sum

We assume that our application will revenue 55,000$ in it's first year, 65,000$ in it's second year and 75,000$ in it's third year. We also assume that we have 10% discount rate.

NPV = Rt/((1 + i)^t) – initial investment

PV1 = 55000/ 1.1 = 50,000

PV2 = 65000/ 1.1^2 = 53,719.01

PV3 = 75000/ 1.1^3 = 56,364.43

PV(total) = 160,083.44

NPV = 160,083.44−100,000 = 60,083.44

ROI calculation:

Total Revenue: 55,000+65,000+75,000=195,000

Net Profit: 195,000−161,000=34,000

ROI = 34,000/161,000 * 100 = 21.1%

The project generates a profitable return on the total investment.

4. Stakeholder Analysis Document

- Stakeholder List: The names and roles of the key stakeholders are as followed:

1. Users – End-users such as private clients that want to move to another apartment and businesses that move their resources.
2. Business Stakeholders –
    a. Product Manager responsible for defining what the product should be and why. They translate business goals into product requirements and make strategic decisions about the app's features.
    b. Project Manager is responsible for how and when the product is delivered. Unlike the Product Manager (who focuses on what to build), the Project Manager focuses on execution.
    c. Company Owners fund, sponsor, or own the product. They define high-level business goals and measure success.
    d. Marketing & Sales Team responsible for bringing users to the app and driving adoption. They shape how the product is seen by customers.
3. Development Team –
    a. Backend Developers responsible for building the server-side logic, databases, and core functionality that power the app.
    b. Frontend build the user interface and ensure the app works smoothly for end users.
    c. QA Testers ensure the app is stable, functional, and free of critical bugs before release.
    d. UX/UI Designers shape the look, feel, and flow of the application.
4. Operations & Maintenance –
    a. DevOps Engineers handle the infrastructure that runs the application. They ensure the system is stable, scalable, and continuously updated.
    b. Technical Support Team handles issues reported by internal teams or by users, usually after the product is already live.
    c. Customer Service handles non-technical support, focusing on helping users with everyday actions inside the app.
5. External Parties –
    a. Map / GPS Providers provide location-based functionality such as navigation, map rendering, and geolocation.
    b. Push Notification Providers enable messaging from the system to users, such as alerts, reminders, and authentication messages.

- Interests and Expectations:

1. Users - End-users such as private clients moving to a new apartment and businesses relocating equipment.

Interests:

- Completing their move quickly, safely, and with minimal stress.
- Finding reliable movers who can handle their belongings with care.
- Getting transparent pricing and clear service details.
- Tracking the move and communicating easily with the service provider.

Expectations:

- A simple, intuitive interface for booking a moving service.
- Accurate time estimates, location tracking, and clear notifications.
- The ability to compare movers based on ratings, availability, and price.
- Fast support in case of issues or last-minute changes.

2. Business Stakeholders

a. Product Manager

Interests:

- Building a product that meets user needs and supports business goals.
- Defining a strong value proposition and competitive advantage.
- Ensuring the product roadmap is realistic and aligned with vision.

Expectations:

- Clear communication from developers and designers.
- Reliable user feedback and analytics to make feature decisions.
- Smooth execution of planned features with predictable delivery timelines.

b. Project Manager

Interests:

- Delivering the project on time, within scope, and within budget.
- Ensuring team coordination and efficient workflow.

Expectations:

- Detailed technical requirements and stable priorities.
- Continual progress reporting from the team.
- Early identification of risks and blockers.

c. Company Owners / Investors

Interests:

- Growing the user base and increasing revenue.

- Achieving a good return on investment.
- Establishing a competitive position in the market.

Expectations:

- Regular performance reports and KPIs.
- A stable product that can scale.
- Positive user satisfaction and strong adoption rates.

d. Marketing & Sales Team

Interests:

- Increasing product visibility and attracting new users.
- Understanding user behavior to improve targeting and campaigns.

Expectations:

- Access to accurate analytics and engagement data.
- A product that is easy to promote—clear features, strong value.
- Consistent updates that support marketing initiatives.

<u>3. Development Team</u>

a. Backend Developers

Interests:

- Building a robust, secure, and scalable backend.
- Implementing clean architecture and efficient logic.

Expectations:

- Clear, consistent requirements from product management.
- Stable APIs, minimal last-minute changes, and good documentation.
- Realistic deadlines and support for testing environments.

b. Frontend Developers

Interests:

- Creating a smooth, responsive, and user-friendly interface.
- Ensuring seamless integration with backend APIs.

Expectations:

- Well-defined UI designs and user flows.
- Reliable backend endpoints and clear API specifications.

- Quick communication with designers and backend developers.

c. QA Testers

Interests:

- Ensuring the application is stable and bug-free.
- Delivering a high-quality product to users.

Expectations:

- Clear acceptance criteria for each feature.
- Access to testing environments, logs, and tools.
- Timely fixes for identified issues.

d. UX/UI Designers

Interests:

- Providing an intuitive, attractive, and efficient user experience.
- Maintaining design consistency across the entire product.

Expectations:

- Early involvement in feature planning.
- Freedom to iterate based on user feedback.
- Proper implementation of design guidelines by developers.

4. Operations & Maintenance

a. DevOps Engineers

Interests:

- Maintaining a stable, scalable, and secure infrastructure.
- Automating deployments and monitoring system performance.

Expectations:

- Predictable release schedules.
- Collaboration from development to ensure deployability.
- Access to logs, metrics, and diagnostic tools.

b. Technical Support Team

Interests:

- Resolving technical issues quickly and efficiently.
- Reducing downtime and user complaints.

Expectations:

- Access to logs, monitoring systems, and documentation.
- Clear escalation procedures for critical bugs.
- Timely responses from development when issues arise.

c. Customer Service

Interests:

- Helping users with non-technical questions and daily use.
- Ensuring customers have a smooth and pleasant experience.

Expectations:

- A stable system with minimal user frustration.
- Clear scripts, guidelines, and product knowledge.
- Notifications when new features or changes are deployed.

5. External Parties

a. Map / GPS Providers

Interests:

- Reliable usage of their APIs and adherence to their terms.
- Maintaining performance and availability for integrated services.

Expectations:

- Developers using the APIs correctly and efficiently.
- Compliance with licensing, quota limits, and branding policies.

b. Push Notification Providers

Interests:

- Ensuring reliable delivery of alerts, reminders, and authentication messages.
- Maintaining service integrity and uptime.

Expectations:

- Proper integration with their APIs.
- Secure handling of authentication keys and configurations.
- Reasonable load levels within usage limits.

- Communication Plan:

1. Users

Communication Channels: In-app notifications for booking updates, SMS/push for urgent alerts, in-app chat with movers.
Frequency: Real-time updates for moves and chat, optional weekly/monthly summary notification.

2. Business Stakeholders

a. Product Manager
Channels: Weekly meetings, Slack, analytics dashboards.
Frequency: Weekly syncs, daily communication as needed, monthly KPI reports.

b. Project Manager
Channels: Slack, status meetings.
Frequency: Daily stand-ups, weekly status reports, immediate updates for risks.

c. Company Owners / Investors
Channels: Email reports, strategy meetings, high-level dashboards.
Frequency: Monthly/quarterly meetings, on-demand updates for major events.

d. Marketing & Sales Team
Channels: Analytics dashboards, Slack, coordination meetings.
Frequency: Weekly planning meetings, real-time updates during launches, monthly performance reports.

3. Development Team (Backend, Frontend, QA, UX/UI)

Channels: Slack, daily stand-ups, design reviews.
Frequency: Daily communication, bi-weekly sprint planning/review, continuous updates during development.

4. Operations & Maintenance

a. DevOps Engineers
Channels: Slack, monitoring tools, release meetings.
Frequency: Real-time alerts, weekly coordination meetings, as-needed communication during deployments.

b. Technical Support
Channels: Helpdesk systems (Zendesk/Freshdesk), Slack for escalations.
Frequency: Daily support operations, immediate escalation for critical issues, weekly issue summaries.

c. Customer Service
Channels: In-app chat, CRM tools, email templates.
Frequency: Real-time or same-day responses, daily summaries to management, monthly satisfaction reports.

5. External Parties

a. Map / GPS Providers
Channels: Email alerts, provider dashboards, documentation updates.
Frequency: As needed for API changes, monthly quota checks, immediate communication during outages.

b. Push Notification Providers
Channels: Provider dashboards, email alerts, release notes.

Frequency: Real-time alerts for failures, monthly performance checks, on-demand during configuration changes.

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